PyroScope – Forest Fire Analytics: Statistical Analysis and Visualization of Forest Fire Patterns Using R

A PROJECT REPORT

Submitted by
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BONAFIDE CERTIFICATE

Certified that this project report **"PyroScope – Forest Fire Analytics: Statistical Analysis and Visualization of Forest Fire Patterns Using R"** is the bonafide work of **"Bhumi Jindal (26MCD10071)"** who carried out the project work under my/our supervision. The content and findings presented herein are authentic and reflect the original work conducted during the course of this project.

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ABSTRACT

Forest fires damage vegetation, wildlife and soil, release large amounts of carbon and put human settlements at risk. Fire records alone, however, cannot tell a forest manager when fires are most frequent, how large they usually become, or which weather and fire-weather conditions go with them. This project, **PyroScope – Forest Fire Analytics**, is a statistical analysis and visualization of the *Forest Fires* dataset, which holds 517 fire records with 13 variables: the grid position of the fire, the month and day, four fire-weather indices (FFMC, DMC, DC and ISI), four weather measurements (temperature, relative humidity, wind speed and rain) and the burned area in hectares. The analysis is written in R and delivered as an R Markdown document using ggplot2, dplyr, tidyr, knitr and corrplot.

The data is first loaded, profiled and checked: all 13 columns are complete and contain no missing values. Descriptive statistics are then calculated for every numeric variable, and the records are counted by **month** and by **day of week**. The distribution of the **burned area** is studied with a histogram, the relationship between **temperature, humidity and wind** is shown with scatter plots, average weather is tabulated by month, and a **correlation matrix** is drawn for all numeric variables.

The analysis shows that fire records are strongly concentrated in the late summer: **August (184 records)** and **September (172 records)** together hold 68.9% of all records, followed by March with 54. Sunday has the most records (95) and Wednesday the fewest (54). The burned area is extremely skewed: 247 records (47.8%) show zero burned area, the median is only **0.52 ha**, the mean is **12.85 ha**, and the largest single fire burned **1,090.84 ha**. The ten largest fires account for about half of the total burned area. Temperature and relative humidity are clearly related ($r = -0.53$), but none of the four weather variables has a strong direct correlation with burned area (all $|r| < 0.10$).

The work is published on GitHub together with the dataset and the screenshots. This report describes the scope, background, design, implementation, results and future scope of the project and includes the charts produced by the analysis.

Keywords: Forest Fire Analytics, Exploratory Data Analysis, Burned Area, Fire Weather Index, Correlation Analysis, R, ggplot2, R Markdown.

List of Abbreviations

Abbreviation	Definition
EDA	Exploratory Data Analysis
CSV	Comma-Separated Values
FWI	Fire Weather Index (Canadian Forest Fire Weather Index System)
FFMC	Fine Fuel Moisture Code
DMC	Duff Moisture Code
DC	Drought Code
ISI	Initial Spread Index
RH	Relative Humidity
SD	Standard Deviation
IDE	Integrated Development Environment
GIS	Geographic Information System
ML	Machine Learning
HTML	HyperText Markup Language
UI	User Interface

CHAPTER 1.
INTRODUCTION

Forest fires are among the most damaging natural hazards. They destroy trees and wildlife habitat, degrade soil, pollute the air, and threaten homes and livelihoods near forests. Fire-management agencies record every fire they deal with: when it started, where it was, what the weather and the dryness of the vegetation were like, and how much land was burned. When hundreds of such records accumulate, the data can no longer be understood line by line, yet it holds the answers to questions that forest managers ask every year: In which months do fires cluster? How large does a typical fire become? Are hot, dry and windy days really linked to bigger fires? Exploratory data analysis (EDA) answers such questions by summarising the data with statistics and visualisations.

PyroScope – Forest Fire Analytics applies EDA to the *Forest Fires* dataset, a file of 517 fire records with 13 variables. The project is deliberately built with the open-source R language and an R Markdown document, so that every step – loading, checking, summarising, plotting and correlation analysis – is visible, reproducible and can be re-run on any other fire file with the same structure. It therefore also serves as a practical demonstration of how raw fire records are turned into understandable information.

1.1. Project Scope and Project Planning:

The scope of the project is the exploratory statistical analysis of the forest fire data using R. The analysis covers: (i) loading and profiling of the dataset; (ii) a statistical summary of every numeric variable and a missing-value check; (iii) fire records by month and by day of the week; (iv) the distribution of the burned area; (v) the relationship between temperature, relative humidity and wind speed; (vi) average weather conditions by month; (vii) a correlation analysis of all numeric variables; and (viii) the ten records with the largest burned area. Predictive modelling, spatial mapping of the fires and interactive dashboards are outside the present scope and are listed under future work.

1.2. Client Identification and Recognition of need: -

The primary clients of the analysis are forest-department officials and fire-management planners, who decide when to station crews, when to issue fire-danger warnings and where to spend prevention budgets. Secondary stakeholders are environmental researchers, who study the link between weather and fire, and national-park authorities, who protect the land and the visitors. Their typical needs are: a clear view of how fire records change through the year; an idea of how large fires usually are and how rare the very large ones are; knowing whether temperature, humidity, wind and rain separate small fires from large ones; and charts that can be read without a statistical background. These needs were converted into the objectives in Section 2.6.

1.3. Recognition and Knowledge of relevant contemporary issues: -

Fire seasons have become longer and more severe in many parts of the world, and forest agencies are expected to plan with data rather than intuition. Several contemporary issues are relevant to this project. **Climate and weather:** hot and dry conditions dry out the vegetation and make ignition and spread easier, which is why fire-danger rating systems are built on weather observations. **Skewed outcomes:** most fires stay small while a few become very large; in this dataset a single fire burned 1,090.84 ha, while about half the records show no burned area, so averages must be read with care. **Correlation versus cause:** weak or moderate correlations do not prove cause and effect, and a descriptive analysis cannot predict future fires by itself. **Data quality and reproducibility:** conclusions are reliable only if the data is checked and the steps can be repeated. **Ethics and privacy:** the dataset contains no personal data; it is a public research dataset, but real fire records with exact locations may need to be handled carefully.

1.4. Task Identification: -

- **Data task:** load forestfires.csv with `read.csv()` and inspect its structure, types and statistics.
- **Quality and summary tasks:** check missing values and compute the mean, median, standard deviation, minimum and maximum of each numeric variable.
- **Occurrence tasks:** count the fire records by month and by day of the week and draw bar charts.
- **Burned-area task:** study the distribution of the burned area, including the number of zero-area records, the mean and the median.
- **Weather tasks:** plot temperature against relative humidity and wind speed against temperature, and tabulate the average weather by month.
- **Relationship and publication tasks:** compute and draw the correlation matrix, list the largest fires, document the work and publish it on GitHub.

1.5. Technologies Used: -

Technology	Purpose in the project
R	Core programming language used for the whole analysis.
RStudio	Integrated development environment used to write, run and knit the code.
R Markdown, knitr	Combines code, output and text in one document and renders it to HTML with a floating table of contents.
dplyr, tidyr	Data manipulation: select(), count(), group_by(), summarise(), arrange().
ggplot2	Grammar-of-graphics plotting: bar charts, histogram, scatter plots and trend line.
corrplot	Draws the correlation matrix as a colour-coded heat map.
forestfires.csv	Data source: 517 fire records, 13 columns, covering all twelve months.
Git, GitHub	Version control and hosting of the R Markdown file, HTML output, dataset, README and screenshots.

1.6. Timeline of the project

The project was carried out in six phases, shown in Figure 1 – problem and data understanding, data loading and preparation, descriptive statistics, visual exploration, correlation analysis with insights, and documentation with GitHub publication. Earlier steps were repeated whenever a later step revealed a data issue.

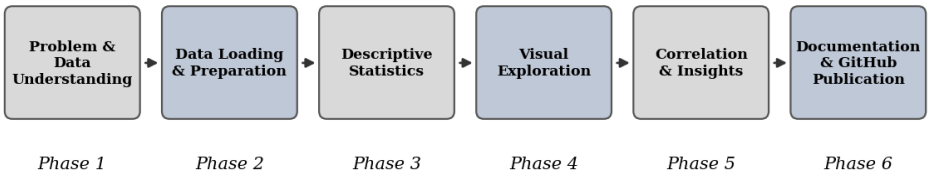


Figure 1: Development lifecycle of the project

CHAPTER 2.
LITERATURE REVIEW/BACKGROUND STUDY

2.1 Timeline of the reported problem:

Forest fires have been studied for as long as forests have been managed, and the main problem reported over time has stayed the same: fires are hard to predict because they depend on many interacting factors. Early fire services relied on lookout towers and experience. In the second half of the twentieth century, fire-danger rating systems were developed that turn weather observations into numerical indices; the Canadian Forest Fire Weather Index System, described by Van Wagner (1987), is the best known and provides the FFMC, DMC, DC and ISI indices used in this dataset. The idea of looking at data visually before building models was set out by John Tukey in his 1977 book on exploratory data analysis. Since the 2000s, data-mining studies such as Cortez and Morais (2007) have tried to predict burned area from weather data, and have found the task difficult because the burned area is extremely skewed towards small values. At every stage the reported problems have been similar: noisy and incomplete records, strongly unbalanced outcomes, and a weak direct link between a single weather variable and fire size.

2.2 Existing solutions:

Fire-danger rating systems such as the Fire Weather Index convert temperature, humidity, wind and rain into moisture codes and spread indices, but they describe danger rather than explain a particular dataset. **Remote-sensing and GIS tools** detect active fires from satellites and map burned areas, but they need specialised data and software. **Machine-learning models** (regression, support vector machines, random forests) try to predict burned area from weather variables, but they need a good understanding of the data first and are hard to interpret. **Spreadsheets** are easy to start with but make repeatable statistics and quality charts harder. **R and RStudio** offer free, reproducible statistics and publication-quality graphics in one environment, and R Markdown keeps the code, results and explanations together, which is the approach chosen for this project.

2.3 Bibliometric analysis:

A review of the books, papers and package documentation listed in the References shows the following.

Key features

- The Fire Weather Index System summarises fuel moisture and fire behaviour in a small number of indices (FFMC, DMC, DC, ISI), which are included in the dataset.
- The grammar of graphics implemented in ggplot2 builds each chart from data, aesthetics and geometric layers, which keeps the code short and consistent.
- R Markdown keeps code, results and explanations together, which makes the analysis easy to review and share.

2.4 Review Summary:

The literature agrees that a sound EDA begins with checking data quality, summarises the data at meaningful levels and uses simple, well-labelled charts for each kind of question: bars for counts, histograms for distributions, scatter plots for relationships and heat maps for correlation matrices. It also warns that fire-size data is strongly skewed and that weather variables alone explain little of it. This project adopts these practices and lists the deeper techniques (log-transformation, spatial analysis, predictive modelling, dashboards) as future work.

2.5 Problem Definition:

To analyse the forest fire dataset using R in order to determine **when fire records occur, how large the burned areas are, and how temperature, humidity, wind and rain relate to each other and to the burned area**, to **identify the months, days and individual fires that stand out**, and to present the findings through clear tables and visualisations that support fire-management decisions.

2.6 Goals/ Objectives:

- To load, profile and validate the forest fire dataset using R.
- To calculate descriptive statistics for all numeric variables and check for missing values.
- To analyse the number of fire records by month and by day of week.
- To study the distribution of the burned area and identify the largest fires.
- To explore the relationships between temperature, relative humidity, wind speed and the burned area using scatter plots and a correlation matrix.
- To publish the R Markdown file, HTML output, dataset and screenshots on GitHub so that the work can be reproduced.

CHAPTER 3.
DESIGN FLOW/PROCESS

The design follows the standard pipeline of a data-analysis project: acquire the data, prepare it, analyse it, visualise the results and communicate them. Each stage uses a small number of dplyr, ggplot2 and base R functions, so that the whole analysis stays easy to follow. The design process moved from identifying the features of the analysis, through constraints and selection, to the final workflow and implementation plan described below.

3.1 Evaluation & Selection of Features: -

Candidate features were evaluated for their usefulness to a fire manager and for the effort involved. Features marked "Included" form the implemented analysis; the others are deferred.

Feature	Reason	Priority	Decision
Data loading and structure	Understand variables, types and size	High	Included
Missing-value check	Reliable results need complete data	High	Included
Descriptive statistics	Summarise every numeric variable	High	Included
Monthly and daily counts	Show when fire records cluster	High	Included
Burned-area distribution	Show how fire sizes vary	High	Included

Feature	Reason	Priority	Decision
Weather relationships and monthly averages	Link temperature, humidity and wind	Medium	Included
Correlation analysis, top-10 fires	Find associations and extreme events	Medium	Included
Spatial map of X, Y grid	Show where fires occur	Medium	Future
Predictive model, interactive dashboard	Advanced analytics and presentation	Low	Future

3.2 Design Constraints: -

- **Data:** a single historical CSV file (no live feed); each row is one fire record, and no year or exact date is given, only month and day of week.
- **Technical:** the analysis must run in standard R with RStudio on an ordinary laptop, using only a few well-known packages.
- **Analytical:** the burned area is extremely skewed, so the mean must be read together with the median, and a correlation is treated as association, not proof of cause.
- **Privacy:** the file holds no personal data; the X and Y values are coarse grid coordinates rather than exact locations.
- **Resource and time:** a single-developer academic project, so static charts and an HTML report are used instead of a full dashboard.
- **Usability:** every chart must have a title and labelled axes so that it can be read without explanation.

3.3 Analysis of Features and finalization subject to constraints:

Considering the constraints, all analyses that depend only on the variables already present in the file were finalised, because they need no extra data and no extra tools beyond the listed R packages. Spatial mapping and predictive modelling were deferred because they need additional tools, a log-transformed target and more time. Static ggplot2 charts were preferred to a dashboard because they can be embedded directly in the knitted HTML file, this report and the GitHub repository.

3.4 Design Flow:

Figure 2 shows the flow of the analysis. The forest fire CSV file is read into an R data frame; its structure is inspected and missing values are checked; descriptive statistics are calculated; the records are counted and averaged by month and day; the results are plotted with ggplot2 and corrplot; and the R Markdown file, data and screenshots are published on GitHub.

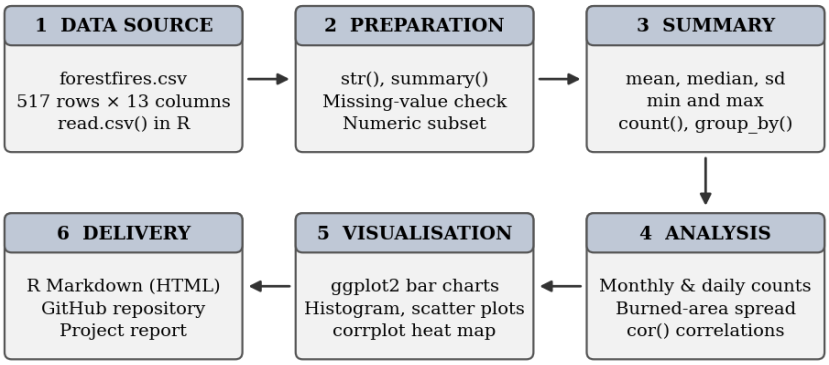


Figure 2: Workflow and architecture of the analysis pipeline

The structure of the input data is given in Table 3.1. All 13 variables are original columns of the file; no new column is added during the analysis.

Table 3.1: Structure of the forest fire dataset

Variable	Type	Description
X	Integer	X-axis position of the fire on the park map grid (1 to 9)

Variable	Type	Description
Y	Integer	Y-axis position of the fire on the park map grid (2 to 9)
month	Text	Month of the year (jan to dec)
day	Text	Day of the week (mon to sun)
FFMC	Decimal	Fine Fuel Moisture Code of the FWI system (18.7 to 96.2)
DMC	Decimal	Duff Moisture Code of the FWI system (1.1 to 291.3)
DC	Decimal	Drought Code of the FWI system (7.9 to 860.6)
ISI	Decimal	Initial Spread Index of the FWI system (0.0 to 56.1)
temp	Decimal	Temperature in degrees Celsius (2.2 to 33.3)
RH	Integer	Relative humidity in per cent (15 to 100)
wind	Decimal	Wind speed in km/h (0.4 to 9.4)
rain	Decimal	Outside rain in mm/m² (0.0 to 6.4)
area	Decimal	Burned area of the forest in hectares (0.00 to 1,090.84)

3.5 Design Selection: -

A script-based, package-driven design was selected: base R and dplyr for data handling, ggplot2 and corrplot for charts, and R Markdown for combining code and results. Compared with a spreadsheet or a BI tool, it is free, transparent and reproducible. The chart type was chosen to match each question: a **bar chart** for counts by month and by day, a **histogram** for the distribution of the burned area, **scatter plots** with a fitted line for relationships between two weather variables, and a **colour-coded correlation matrix** for all numeric variables together.

3.6 Implementation plan/methodology

Development was incremental: the data was loaded and checked first, then each analysis was written, run and visualised before the next one was started.

CHAPTER 4.
RESULTS ANALYSIS AND VALIDATION

4.1 Implementation of the solution:

The analysis was implemented in the R Markdown file *PyroScope.Rmd* and knitted to *PyroScope.html*. The sections below present the output of each module together with its interpretation. The burned area is measured in hectares (ha), temperature in degrees Celsius, relative humidity in per cent and wind speed in km/h.

Data loading and profiling. The file was read with `read.csv()`, giving 517 records and 13 columns. The `str()` output showed that eight columns are decimal variables (FFMC, DMC, DC, ISI, temp, wind, rain and area), three are integers (X, Y and RH), and month and day are text. The check `colSums(is.na())` returned zero missing values in every column. As an additional check while preparing this report, four rows were found to be exact duplicates of other rows; they were kept, as in the original analysis, because they do not change any conclusion. Table 4.1 gives an overview of the data and Table 4.2 the statistical summary of the numeric columns.

Table 4.1: Overview of the dataset

Item	Value
Records / columns	517 / 13
Numeric / text columns	11 / 2 (month and day)
Missing values	0
Records with zero burned area	247 (47.8%)
Records with positive burned area	270 (52.2%)
Total burned area	6,642.05 ha
Largest single burned area	1,090.84 ha

Table 4.2: Statistical summary of numeric columns

Variable	Mean	Std. dev.	Minimum	Median	Maximum
X	4.67	2.31	1.0	4.0	9.0
Y	4.30	1.23	2.0	4.0	9.0
FFMC	90.64	5.52	18.7	91.6	96.2
DMC	110.87	64.05	1.1	108.3	291.3
DC	547.94	248.07	7.9	664.2	860.6
ISI	9.02	4.56	0.0	8.4	56.1
temp	18.89	5.81	2.2	19.3	33.3
RH	44.29	16.32	15.0	42.0	100.0
wind	4.02	1.79	0.4	4.0	9.4
rain	0.02	0.30	0.0	0.0	6.4
area	12.85	63.66	0.00	0.52	1,090.84

The mean burned area (12.85 ha) is far above the median (0.52 ha) and the standard deviation (63.66 ha) is five times the mean, which shows that a few very large fires pull the average up. Rain is almost always zero: only 8 of the 517 records have any rain, and the maximum is 6.4 mm. The FFMC index is high in most records (median 91.6), reflecting dry fine fuels in the fire season.

Fire records by month. The records were counted by month and plotted as a bar chart (Figure 3). Fire records are overwhelmingly concentrated in the late summer: **August has 184 records and September 172**, together 68.9% of the dataset. March is a distant third with 54 records and July follows with 32. November (1), January (2) and May (2) have almost no records. The monthly table (Table 4.3) also shows that the average temperature is highest in July (22.1 °C) and lowest in December (4.5 °C), that the average relative humidity is lowest in November (31.0%) and October (37.5%), and that the highest average wind speed is recorded in December (7.64 km/h, from only 9 records). The months with only one or two records give unreliable averages and should be read with caution.

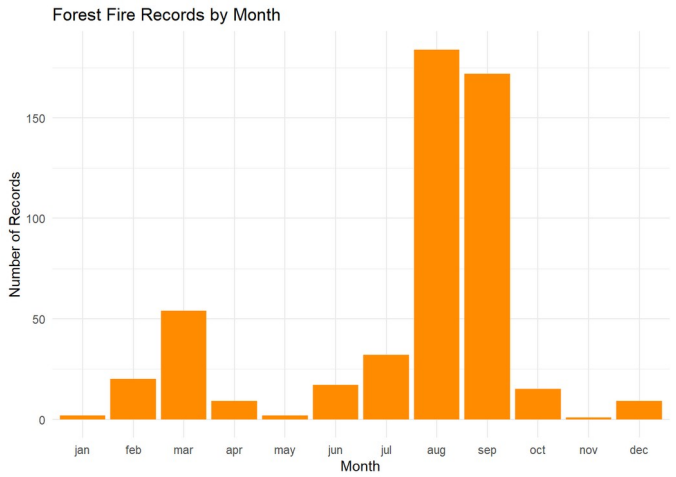


Figure 3: Forest fire records by month

Table 4.3: Fire records and average weather by month

Month	Records	Share	Avg temp (°C)	Avg RH (%)	Avg wind (km/h)	Burned area (ha)
January	2	0.4%	5.25	89.00	2.00	0.00
February	20	3.9%	9.64	55.70	3.76	125.50
March	54	10.4%	13.08	40.00	4.97	235.26
April	9	1.7%	12.04	46.89	4.67	80.02
May	2	0.4%	14.65	67.00	4.45	38.48
June	17	3.3%	20.49	45.12	4.14	99.30
July	32	6.2%	22.11	45.12	3.73	459.83

Month	Records	Share	Avg temp (°C)	Avg RH (%)	Avg wind (km/h)	Burned area (ha)
August	184	35.6%	21.63	45.49	4.09	2,297.99
September	172	33.3%	19.61	42.84	3.56	3,086.13
October	15	2.9%	17.09	37.47	3.46	99.57
November	1	0.2%	11.80	31.00	4.50	0.00
December	9	1.7%	4.52	38.44	7.64	119.97
Total	**517**	**100.0%**	–	–	–	**6,642.05**

September and August also account for most of the burned area: 3,086 ha (46.5%) and 2,298 ha (34.6%), so the two months together hold 81.1% of the total burned area.

Fire records by day of week. Figure 4 shows that the records are spread across the week without a sharp peak. Sunday has the most records (95), followed by Friday (85), Saturday (84) and Monday (74), while Tuesday (64), Thursday (61) and Wednesday (54) have fewer. Friday to Monday together contain 338 records (65.4%). The ggplot2 chart lists the days alphabetically (fri, mon, sat, sun, thu, tue, wed), which is the default order of a text variable, so the chart should be read day by day rather than as a weekly sequence. The data cannot show whether the weekend pattern is due to human activity or only to chance.

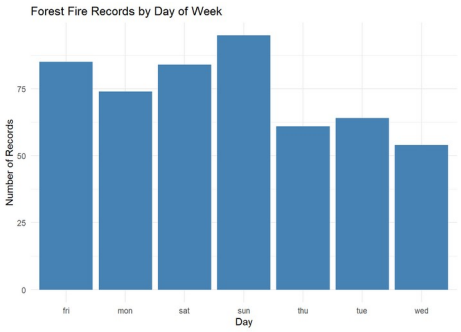


Figure 4: Forest fire records by day of week

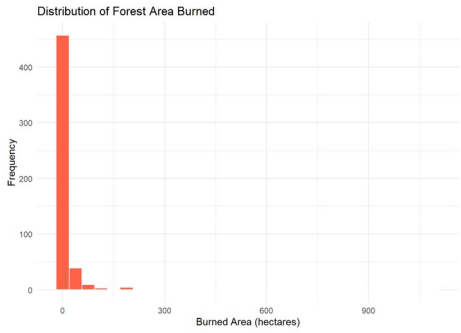


Figure 5: Distribution of the burned area

Distribution of the burned area. The histogram in Figure 5 has one very tall bar at zero and a long thin tail to the right. Of the 517 records, 247 (47.8%) have zero burned area and 270 have a positive value; 422 records (81.6%) are below 10 ha and only 11 exceed 100 ha. The quartiles are 0 ha, 0.52 ha and 6.57 ha (Q1, median, Q3), and the 95th percentile is only 48.7 ha. In contrast, the maximum is 1,090.84 ha. The ten largest fires burned 3,346.9 ha, which is 50.4% of the total burned area of 6,642.05 ha, so half of all burned land comes from about 2% of the records. The mean exceeds the median by a wide margin, which is the signature of a strongly right-skewed distribution.

Temperature, humidity and wind. Figure 6 plots relative humidity against temperature with a fitted straight line. The line slopes downwards: on average the humidity falls by about 1.5 percentage points for each degree of temperature (RH ≈ 72.3 – 1.48 × temp), and the correlation is –0.53, i.e. temperature explains about 28% of the variation in humidity. The points are widely scattered around the line, so the relation is moderate rather than strict. Figure 7 shows wind speed against temperature. The points form a broad cloud with only a faint downward tilt (r = –0.23): windier days are slightly cooler, but wind and temperature are largely independent of one another.

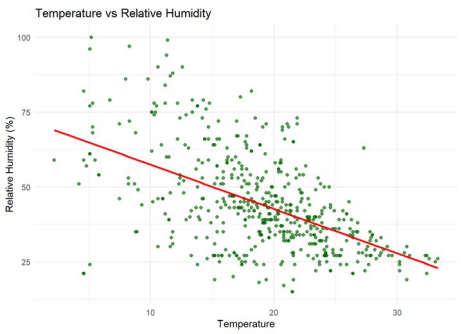


Figure 6: Temperature vs relative humidity

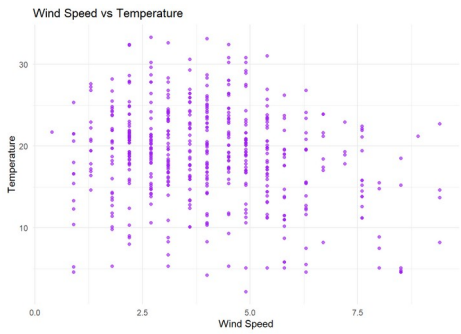


Figure 7: Wind speed vs temperature

Correlation analysis. The correlation matrix of all 11 numeric variables was calculated with `cor()` and drawn with `corrplot` (Figure 8). The upper triangle shows the strongest links between the variables themselves (Table 4.4): the Duff Moisture Code and the Drought Code move together (r = 0.68), because both

measure the dryness of deeper fuel layers; FFMC and ISI are linked (0.53), because ISI is partly computed from FFMC; temperature is negatively related to relative humidity (–0.53) and positively to the drought and moisture codes (DC 0.50, DMC 0.47). The X and Y grid positions are related (0.54) because the fires are not spread evenly over the map.

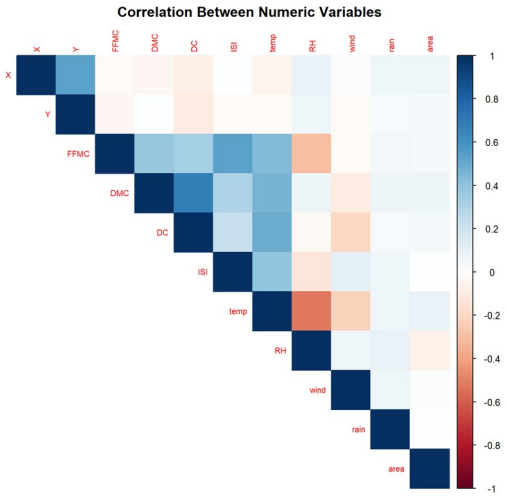


Figure 8: Correlation matrix of the numeric variables

The bottom row of the matrix is the most important for this project: the correlation of each variable with the burned area. All coefficients are close to zero. Temperature has the highest value (0.098), relative humidity is slightly negative (–0.076), and wind (0.012) and rain (–0.007) show practically no linear association. Even the best of the fire-weather indices, DMC, reaches only 0.07. The weather measured on the day therefore does not explain how large a fire becomes in a simple linear way.

Table 4.4: Selected correlation coefficients

Pair of variables	r	Reading
DMC and DC	0.68	Strong positive: both measure deep-layer dryness
X and Y	0.54	Positive: spatial pattern of the fire locations
FFMC and ISI	0.53	Positive: ISI is partly built from FFMC
temp and RH	–0.53	Moderate negative: hotter days are drier
temp and DC	0.50	Moderate positive
wind and temp	–0.23	Weak negative
temp and area	0.098	Very weak positive
RH and area	–0.076	Very weak negative
wind and area	0.012	Practically none
rain and area	–0.007	Practically none

Records with the largest burned areas. Table 4.5 lists the ten largest fires. Nine of the ten occurred in August or September, and the two largest – 1,090.84 ha in September and 746.28 ha in August – burn 27.7% of all land in the dataset between them. The weather on those days was not extreme: the temperatures were 25.1 °C and 27.5 °C, the relative humidity 27%, and the wind speed 4.0 and 4.9 km/h, which is ordinary summer weather. One of the ten largest fires, 278.53 ha, even occurred at a humidity of 57%. This again shows that weather alone does not distinguish large fires from small ones.

Table 4.5: Top 10 records by burned area

Rank	Month	Day	Temp (°C)	RH (%)	Wind	Area (ha)
1	sep	sat	25.1	27	4.0	1,090.84
2	aug	thu	27.5	27	4.9	746.28
3	jul	mon	22.6	57	4.9	278.53
4	sep	tue	18.8	40	2.2	212.88
5	sep	sat	18.2	46	1.8	200.94

Rank	Month	Day	Temp (°C)	RH (%)	Wind	Area (ha)
6	aug	sun	19.6	41	5.8	196.48
7	aug	wed	26.2	36	4.5	185.76
8	aug	sat	21.9	42	2.2	174.63
9	sep	sat	17.7	25	3.1	154.88
10	sep	tue	24.3	36	3.1	105.66

4.2 Summary of findings and validation:

Key findings

- The dataset holds **517** fire records and **13** variables with **no missing values**; **247 records (47.8%)** have zero burned area and **270** have a positive burned area.
- Fire records are strongly seasonal: **August (184) and September (172) hold 68.9%** of all records and 81.1% of the burned area; March (54) is a distant third.
- The burned area is extremely skewed: the **median is 0.52 ha**, the **mean 12.85 ha** and the **maximum 1,090.84 ha**; the ten largest fires account for about **half** of all burned land.
- Temperature and humidity are moderately related (**r = –0.53**), but temperature, humidity, wind and rain each have a correlation with burned area **below 0.10** in absolute value.
- Sunday (95 records) and Friday (85) have the most records and Wednesday (54) the fewest, but the differences between days are modest.

Validation

The figures in this report were checked against the dataset: the counts, means, medians and extremes were recomputed from the numeric columns, the monthly counts were reconciled with the total number of records, and the burned area by month was reconciled with the overall total. The checks are summarised below.

Table 4.6: Test summary

No.	Test case	Expected result	Result
1	Load forestfires.csv with read.csv()	517 rows × 13 columns loaded	Pass
2	Check for missing values	Zero missing values in every column	Pass
3	Check for duplicate rows (extra check)	Count reported; rows retained	4 found, noted
4	Statistical summary of numeric columns	Mean, median, SD, min and max for 11 variables	Pass
5	Reconcile monthly counts	The 12 monthly counts sum to 517	Pass
6	Reconcile burned area by month	Monthly areas sum to 6,642.05 ha	Pass
7	Correlation matrix	Symmetric, diagonal equal to 1, values from –1 to 1	Pass
8	Identify the largest fires	Top 10 sorted with arrange(desc(area))	Pass
9	Draw all charts	Titles, axes and labels displayed correctly	Pass

CHAPTER 5.
CONCLUSION AND FUTURE WORK

5.1 Conclusion: -

This project achieves its objectives: the forest fire dataset was loaded, checked and summarised with R, and its fire records, burned area and weather variables were analysed by month, day of week, distribution and correlation and presented in clear charts and tables. The analysis turned 517 raw fire records into a set of practical findings. Fire records are concentrated in August and September; most fires burn little or nothing while a handful burn very large areas; temperature and humidity are linked but neither explains the size of a fire in a simple way; and the largest fires happened under ordinary summer weather. The project also gave hands-on experience of the full life cycle of a data-analysis project – from data understanding and checking to visualisation, documentation and publication on GitHub.

Based on these findings, a forest-management team could (i) concentrate patrols, equipment and public warnings on the late-summer months; (ii) treat the very large fires as a separate, rare class that needs its own study rather than relying on the average fire size; and (iii) avoid judging fire danger from a single weather measurement such as temperature. These points are observations from one dataset and should be tested on more data before they are used for decisions.

The present version is a foundation rather than a finished product. The analysis is descriptive and uses linear correlation only; the burned area is not log-transformed; the data has no year, so trends over time are not shown; the grid position is not mapped; the charts are static; and the dataset is a single sample from one region. Observed relationships do not establish the causes of forest fires or predict future events. These limitations form the basis of the future scope.

5.2 Future Scope: -

- Apply a logarithmic transformation (log of area + 1) to the burned area and repeat the correlation and distribution analysis on the transformed values.
- Analyse the X and Y grid positions with a spatial map to find areas of the park where fires are more frequent or larger.
- Build and compare predictive models (linear and logistic regression, random forest, support vector machine) to predict whether a fire will be small or large.
- Study the Fire Weather Index variables (FFMC, DMC, DC, ISI) in more depth, including their interaction with season and weather.
- Compare fire records and burned area by season and by weekday with statistical tests, instead of visual comparison alone.
- Build an interactive dashboard in Shiny, Power BI or Plotly so that managers can filter by month, weather and fire size, and connect the analysis to a database instead of a CSV file.

APPLICATION CODE OF PROJECT

The main code chunks of the R Markdown file are given below. The complete file, *PyroScope.Rmd*, the knitted *PyroScope.html* and the dataset are available in the GitHub repository. In the original file the data is read from an absolute path on the author's computer; here it is shortened to "forestfires.csv" for clarity.

1. Loading libraries and the data

```
library(ggplot2); library(dplyr); library(tidyr); library(knitr); library(corrplot)

forest_data <- read.csv("forestfires.csv")
cat("Number of records:", nrow(forest_data), "\n")
str(forest_data)
summary(forest_data)
```

2. Statistical summary and missing values

```
numeric_data <- forest_data %>% select(where(is.numeric))

statistical_summary <- data.frame(
  Variable = names(numeric_data),
  Mean = sapply(numeric_data, mean, na.rm = TRUE),
  Median = sapply(numeric_data, median, na.rm = TRUE),
  Standard_Deviation = sapply(numeric_data, sd, na.rm = TRUE),
  Minimum = sapply(numeric_data, min, na.rm = TRUE),
  Maximum = sapply(numeric_data, max, na.rm = TRUE))
knitr::kable(statistical_summary, digits = 2)

missing_values <- colSums(is.na(forest_data))
```

6. Correlation analysis, largest fires and insights

```
correlation_matrix <- cor(numeric_data, use = "complete.obs")
corrplot(correlation_matrix, method = "color", type = "upper", tl.cex = 0.7,
         title = "Correlation Between Numeric Variables", mar = c(0, 0, 2, 0))
```

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USER MANUAL

A. Setting up the project

1. Install R and RStudio, then install the required packages in the R console: `install.packages(c("ggplot2", "dplyr", "tidyr", "knitr", "corrplot", "rmarkdown"))`.
2. Download or clone the GitHub repository and keep forestfires.csv in the same folder as PyroScope.Rmd.
3. Open PyroScope.Rmd in RStudio and change the file path in the read.csv() call to "forestfires.csv" (or to the full path on your computer).

B. Using the R Markdown file

1. Click **Knit → Knit to HTML**. The data is loaded, the statistical summary is shown and the file PyroScope.html is created.
2. Check the data-quality table: every column should show zero missing values.
3. Read the monthly and daily charts (Figures 3 and 4) to see when fire records occur.
4. Read the burned-area histogram (Figure 5) and the top-10 table to see how fire sizes vary.
5. Read the scatter plots (Figures 6 and 7) and the correlation heat map (Figure 8) to see how the weather variables relate to each other and to the burned area.
6. To analyse another fire file, replace the CSV with a file that has the same column names and knit the document again.

GITHUB REPOSITORY

The complete project (R Markdown file, knitted HTML, dataset, README and screenshots) is published on GitHub at: https://github.com/jindalbhumi125-dev/PyroScope-Forest-Fire-Analytics

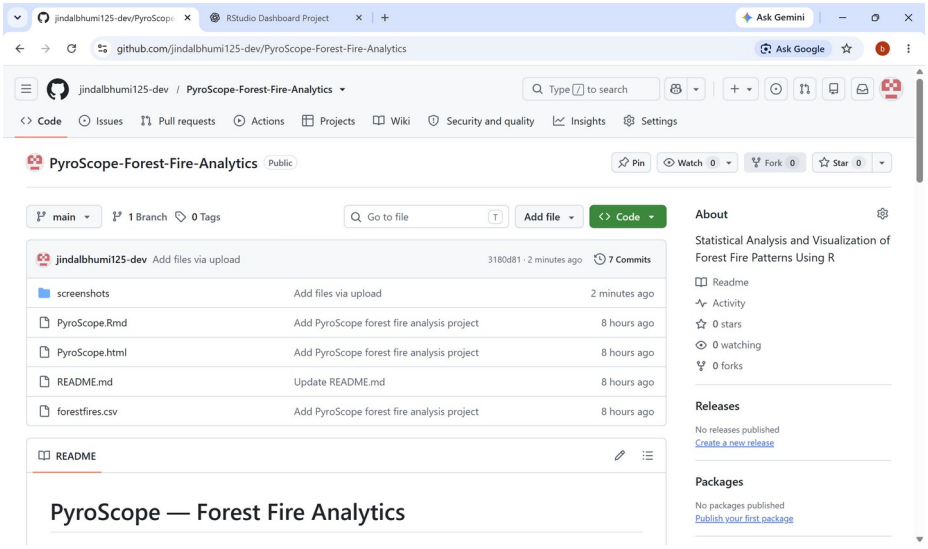


Figure 9: GitHub repository of the project